Abstract

The quantum measurement problem asks why a definite outcome occurs when quantum mechanics predicts only a probability distribution. We present an observer-independent criterion for wave function collapse: an interaction constitutes a genuine actualization event if and only if it produces an irreversible increase in quantum relative entropy $\Delta S(\rho\|\sigma_0) > 0$ with respect to the Minkowski vacuum state σ_0. This criterion is basis-independent, non-perturbative, and does not require an observer, a Heisenberg cut, or a preferred foliation. In weakly coupled quantum field theory it coincides with on-shell emission ($p^\mu p_\mu = m^2c^2$), recovering the standard perturbative picture as a limiting case. The framework resolves the measurement problem by dissolving it: the question "why does a definite outcome occur?" is answered by "because an irreversible entropy-producing event occurred," which is a physical fact about the interaction, not a postulate about observers. We derive three experimental predictions that discriminate this framework from all current interpretations of quantum mechanics: (1) a finite actualization propagation velocity bounded by the Lieb-Robinson speed; (2) an environment-dependent transition duration scaling as $\tau_{\text{act}} \sim \hbar/\Delta E_{\text{env}}$; and (3) a sharp, measurable reversibility threshold above which quantum transitions cannot be undone regardless of intervention. All three are absent from dynamical collapse models, decoherence-only accounts, and Everettian approaches. The Rindler thermal bath satisfies $\Delta S(\rho\|\sigma_0) = 0$ under modular flow, showing that a uniformly accelerating observer detects no actualization events in the Minkowski vacuum—a result formally verified in Lean 4/Mathlib.

Wave Function Collapse as Irreversible Entropy Production: An Observer-Independent Criterion with Three Experimental Predictions

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1 Introduction

The measurement problem is the most persistent foundational puzzle in quantum mechanics [1]. The Schrödinger equation is unitary and linear: it evolves superpositions into larger superpositions without selecting an outcome. Yet every experiment produces a definite outcome. Reconciling these two facts requires either a modification of the dynamics (dynamical collapse [2, 3]), a reinterpretation of outcomes (Everett [4], relational QM [5]), or an additional ontological element (pilot wave [6]).

None of these approaches offers an observer-independent physical criterion for *when* collapse occurs. Dynamical collapse models introduce new free parameters (the GRW collapse rate λ, the correlation length r_C) and have been constrained but not verified by experiment. Everettian approaches deny that collapse occurs at all. Relational QM grounds collapse in observations but does not specify what constitutes an observation. Decoherence accounts correctly identify environmental entanglement as the mechanism for apparent classicality but, as Joos and Zeh [7] and later Zurek [8] both acknowledged, decoherence does not by itself select a definite outcome.

We propose a criterion that is physical rather than observer-relative, non-perturbative rather than tied to Feynman diagrams, and falsifiable rather than merely interpretive. The central claim is stated in Section 2. Sections 3–4 develop its consequences: the dissolution of the measurement problem, the Unruh consistency check, and three discriminating experimental predictions. Section 5 positions the framework against competing approaches.

2 The Actualization Criterion

2.1 Setup

Let $\mathcal{H}$ be the Hilbert space of a quantum system S coupled to an environment E. The composite state evolves under the Schrödinger equation:

$$|\Psi(t)\rangle = \alpha|x_1\rangle|\varepsilon_1(t)\rangle + \beta|x_2\rangle|\varepsilon_2(t)\rangle, \tag{1}$$

where $|\varepsilon_1\rangle$, $|\varepsilon_2\rangle$ are environmental states correlated with the system branches. The reduced density matrix of S is $\rho_S(t) = \mathrm{Tr}_E\big[|\Psi(t)\rangle\langle\Psi(t)|\big]$. The coherences are suppressed by the overlap $\langle\varepsilon_2(t)|\varepsilon_1(t)\rangle \to 0$ as the environment records which-branch information.

Let σ_0 denote the Minkowski vacuum state of the field. The quantum relative entropy of ρ_S with respect to σ_0 is [10, 11]

$$S(\rho_S\|\sigma_0) = \mathrm{Tr}\big[\rho_S(\ln\rho_S - \ln\sigma_0)\big]. \tag{2}$$

2.2 The criterion

Actualization Criterion. An interaction between system S and environment E constitutes a genuine *actualization event*—a wave function collapse in the ontologically real sense—if and only if it produces an irreversible increase

$$\Delta S(\rho_S\|\sigma_0) > 0 \tag{3}$$

in the quantum relative entropy of S with respect to the vacuum state σ_0.

Several features of this criterion are immediate.

Basis-independence. The relative entropy (2) is defined at the level of the density matrix and does not depend on the choice of computational basis.

Observer-independence. The criterion refers to a physical property of the interaction ($\Delta S > 0$), not to any observer's record or knowledge. The vacuum state σ_0 is a physical state of the field, not a subjective reference.

Non-perturbative. The criterion is formulated in terms of the density matrix and applies regardless of coupling strength, extending beyond the domain of Feynman perturbation theory.

Irreversibility as the key. The condition is $\Delta S > 0$, not merely $\Delta S \neq 0$. A reversible interaction (such as a coherent scattering event that does not leave a lasting environmental record) satisfies $\Delta S = 0$ and is not an actualization event. This distinguishes the criterion from entropy production in non-equilibrium thermodynamics: not all entropy-producing processes constitute actualizations, but all actualizations are entropy-producing.

2.3 On-shell condition as perturbative limit

In weakly coupled quantum field theory, an interaction in which a real, on-shell excitation satisfying $p^\mu p_\mu = m^2c^2$ propagates into the asymptotic environment is a *sufficient* condition for (3). An on-shell emission creates environmental states that are asymptotically orthogonal and generates a lasting, redundant causal record. Purely virtual (off-shell) exchange satisfies $\Delta S = 0$ and is not an actualization. The on-shell condition is therefore the perturbative implementation of the entropic criterion, not a separate postulate.

3 Resolution of the Measurement Problem

3.1 The puzzle dissolved

The measurement problem asks: why does the system take on a definite value when quantum mechanics predicts only a probability distribution?

The answer within this framework is: because an actualization event satisfying (3) has occurred. The transition from superposition to definite outcome is not a postulate; it is the physical fact that the interaction produced an irreversible increase in relative entropy. The question is dissolved rather than solved: asking "why did collapse occur?" is equivalent to asking "why did this interaction produce a permanent environmental record?"—which is a question about interaction physics, not about the foundations of quantum mechanics.

3.2 Schrödinger's cat

The cat thought experiment is often presented as showing that quantum superpositions can persist at macroscopic scales. Within this framework, a real cat is continuously actualizing: it metabolizes, moves, and emits heat—a dense internal network of on-shell interactions that pins it firmly in the classical regime. The genuine locus of quantum indeterminacy is the isotope: a small, isolated nucleus whose decay satisfies (3) at the moment of emission. The cat is never in superposition because it is a densely coupled actualization network, not a microscopic superposition. The paradox dissolves.

3.3 The Heisenberg cut is unnecessary

Decoherence-based accounts of measurement require a Heisenberg cut—a boundary at which quantum mechanics hands off to classical physics. The criterion (3) removes the need for such a cut. The classical regime is the dense-graph limit of the same dynamics: as the environmental coupling strength increases, $\langle \varepsilon_2(t) | \varepsilon_1(t) \rangle \to 0$ rapidly, and (3) is satisfied on timescales far shorter than any observable interval. There is no boundary between quantum and classical regimes, only a gradient in the local actualization rate.

3.4 Unruh consistency

A uniformly accelerating observer (Rindler observer) perceives the Minkowski vacuum as a thermal bath at temperature $T_U = \hbar a/(2\pi c k_B)$ [9]. A potential paradox arises: does the Rindler thermal bath trigger actualization events in the Minkowski vacuum?

The criterion (3) resolves this cleanly. The Rindler thermal bath is the Minkowski vacuum expressed in the Rindler algebra; under modular flow, the relative entropy satisfies $\Delta S(\rho \| \sigma_0) = 0$ [12]. No actualization event occurs. Both the inertial observer (who sees $|0_M\rangle$) and the Rindler observer (who sees a thermal bath) agree: there are no physical snap events in the Minkowski vacuum. This result has been formally verified in Lean 4/Math-

lib as the theorem `rindler_stationarity` in `RA_AQFT_Proofs_v10.lean`: the Rindler density matrix ρ_β satisfies $S(\rho_\beta \| \sigma_0) = 0$ under modular flow.

4 Three Experimental Predictions

The criterion (3) makes three quantitative predictions that are absent from all current competing frameworks. Table 1 summarizes the discriminating power against four alternative approaches.

4.1 Prediction 1: Finite actualization propagation velocity

The criterion (3) requires a physical mechanism to propagate from the locus of the interaction to all regions whose superposition is terminated. This mechanism is the spread of environmental entanglement, which is bounded by the Lieb-Robinson speed [13].

Prediction. In a spatially extended quantum many-body system with nearest-neighbor coupling J and lattice spacing ℓ, the decoherence of a distant site B following an actualization event at site A exhibits a time delay

$$\Delta t \geq \frac{|r_A - r_B|}{v_{\mathrm{LR}}}, \qquad v_{\mathrm{LR}} \sim \frac{J\ell}{\hbar}, \tag{4}$$

before the coherences at B begin to collapse. Instantaneous collapse models (GRW, CSL) predict $\Delta t = 0$. Everettian and decoherence-only accounts predict that "collapse" at B is merely apparent and has no physical propagation velocity. The prediction (4) is therefore *uniquely* discriminating: a measured propagation delay at the Lieb-Robinson speed would confirm the physical, finite-velocity spread of actualization.

Experimental access. Ultracold atoms in optical lattices can be initialized in controlled superpositions and the spread of decoherence mapped in real time [14]. Recent experiments with superconducting qubit chains [15] have demonstrated Lieb-Robinson light-cone dynamics. Extracting a decoherence propagation speed from such data would directly test (4).

4.2 Prediction 2: Environment-dependent transition duration

The actualization timescale is set by how long it takes the environmental entanglement to become irreversible, which is controlled by the environmental coupling strength.

Prediction. The duration of a quantum transition (collapse timescale) scales as

$$\tau_{\mathrm{act}} \sim \frac{\hbar}{\Delta E_{\mathrm{env}}}, \tag{5}$$

where ΔE_{env} is the characteristic energy scale of the environmental interaction driving the entanglement. This is not a property of the system alone; it is governed by the environment.

Experimental support. Minev et al. [16] demonstrated that quantum jumps in superconducting artificial atoms are continuous and governed by the coupling to the measurement apparatus, consistent with (5). Pazourek et al. [17] showed that photoemission durations depend on the local Coulomb environment.

The discriminating test. Systematically varying the environmental coupling g in a controlled superconducting circuit—equivalently, tuning $\Delta E_{\mathrm{env}} \propto g$—should produce a proportional variation in τ_{act}. This is a quantitative, parameter-free prediction: the slope $d\tau_{\mathrm{act}}/d(1/\Delta E_{\mathrm{env}}) = \hbar$ is a universal constant. No free parameters are introduced.

4.3 Prediction 3: Sharp reversibility threshold

The criterion (3) distinguishes two regimes separated by a physically sharp boundary:

- *Pre-actualization:* $\Delta S < \Delta S^*$ (threshold not yet crossed). The transition is reversible in principle; the environmental record is not yet permanent.
- *Post-actualization:* $\Delta S > \Delta S^*$. The causal record is permanent. The transition cannot be reversed regardless of any intervention.

Prediction. There exists a sharp threshold $\Delta S^* > 0$ at which reversibility becomes impossible. Below it, quantum erasure succeeds; above it, no erasure protocol can restore the superposition.

Experimental support. Quantum eraser experiments [18] demonstrate that which-path information can be erased before the entanglement becomes irreversible. Minev et al. [16] showed that quantum jumps can be reversed if the intervention occurs before the jump completes.

The discriminating test. Dynamical collapse models (GRW, CSL) posit a stochastic collapse with no sharp threshold—the collapse probability is nonzero at all times, with no clean boundary. Everettian accounts deny the threshold exists. The present criterion predicts a *specific, measurable* point of no return: the entropy production ΔS crossing a critical value determined by the interaction Hamiltonian and the vacuum state geometry. Mapping this threshold in a superconducting qubit system or optical cavity would constitute a direct measurement of the actualization criterion itself.

5 Discussion

5.1 Positioning against competing frameworks

vs. GRW/CSL. Dynamical collapse models [2, 3] modify the Schrödinger equation by adding stochastic terms. They introduce two free parameters (λ, r_C) not determined by any deeper principle. The entropic criterion introduces no free parameters: the threshold is determined entirely by the relative entropy with respect to the vacuum state. Furthermore, GRW/CSL predict no Lieb-Robinson propagation delay and no sharp threshold in the entropic sense.

vs. Everett. Everettian QM [4] denies that collapse occurs. All branches co-exist; measurement reveals pre-existing correlations. The criterion (3) identifies a physical process that distinguishes one branch as actual: the irreversible entropy production. The Everettian cannot explain why $\Delta S > 0$ picks out one branch, because the Everettian framework denies that any selection occurs. The three predictions in Section 4 are therefore directly falsifying for Everett.

vs. Relational QM. Rovelli's relational QM [5] makes collapse relative to observers. The criterion (3) is observer-independent: the relative entropy with respect to the vacuum σ_0 is a physical quantity, not relative to any observer. The Unruh result (Section 3) demonstrates this: both inertial and Rindler observers agree on the $\Delta S = 0$ result.

vs. decoherence-only. Decoherence correctly identifies environmental entanglement as the mechanism for apparent classicality. The criterion adds the crucial step: it specifies *which* environmental entanglements constitute genuine actualizations (those for which $\Delta S > 0$ is *irreversible*). The sharp reversibility threshold (Prediction 3) is the key discriminator: decoherence accounts predict no such threshold, because they deny that genuine collapse occurs.

5.2 Ontic randomness

The framework implies irreducibly ontic randomness. In a two-state superposition $\alpha|x_1\rangle + \beta|x_2\rangle$, the Born rule probabilities $|\alpha|^2$, $|\beta|^2$ are the prior distribution over possible actualization locations. When the actualization occurs at x_1 rather than x_2, this is not determined by any prior fact about the system's history. The actualization event is the unique, ontologically primitive first link of a new causal chain. This is not a failure of the framework to explain randomness; it is the framework's positive claim that the universe is genuinely open.

5.3 Lean 4 formal verification

The core finite-dimensional algebraic structure of the criterion has been formally verified in Lean 4/Mathlib. Key results include: (i) frame independence—$S(U\rho U^\dagger\|\sigma_0) = S(\rho\|\sigma_0)$ for any unitary U (theorem `frame_independence` in `RA_AQFT_Proofs_v10.lean`); (ii) Rindler stationarity—the Rindler thermal state satisfies $\Delta S = 0$ under modular flow (theorem `rindler_stationarity`); and (iii) the data-processing inequality (monotonicity of relative entropy under CPTP maps). The proof files are available at github.com/jsandeman/Relational-Actualism.

6 Conclusion

We have presented an observer-independent criterion for wave function collapse: the irreversible increase in quantum relative entropy $\Delta S(\rho\|\sigma_0) > 0$ with respect to the vacuum state. The criterion is basis-independent, non-perturbative, and introduces no free parameters. It dissolves the measurement problem by identifying collapse with a physical event (irreversible entropy production) rather than an observer-relative postulate. Three experimental predictions discriminate this framework from all current alternatives: the finite Lieb-Robinson propagation velocity of actualization, the environment-dependent transition duration scaling as $\hbar/\Delta E_{\text{env}}$, and the sharp reversibility threshold ΔS^*. The

Table 1: Discriminating predictions of the entropic actualization criterion against four competing frameworks. ✓ = prediction made; × = prediction absent or contradicted; ∼ = partial overlap. RA = Relational Actualism (this work); GRW/CSL = dynamical collapse; EQM = Everettian many-worlds; RQM = relational QM; Deco. = decoherence-only.

Prediction	RA	GRW/CSL	EQM	RQM	Deco.
Finite LR propagation delay ($\Delta t \geq \lvert r_A - r_B \rvert / v_{\mathrm{LR}}$)	✓	×	×	×	×
Environment-dependent duration ($\tau_{\mathrm{act}} \sim \hbar/\Delta E_{\mathrm{env}}$)	✓	∼	×	×	∼
Sharp reversibility threshold (ΔS^* measurable)	✓	×	×	×	×

Rindler thermal bath is consistent with the criterion: $\Delta S = 0$ under modular flow, formally verified in Lean 4.

The framework is part of a broader programme—Relational Actualism—in which the same irreversibility criterion yields the structural zero of the cosmological constant, a mechanism for the Hubble tension, and the origin of biological complexity from a single ontological commitment [19, 20]. These consequences are developed in companion papers.

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References

[1] J. S. Bell, *Against "Measurement"*, Phys. World **3**, 33 (1990).

[2] G. C. Ghirardi, A. Rimini, and T. Weber, Unified dynamics for microscopic and macroscopic systems, Phys. Rev. D **34**, 470 (1986).

[3] P. Pearle, Combining stochastic dynamical state-vector reduction with spontaneous localization, Phys. Rev. A **39**, 2277 (1989).

[4] H. Everett III, "Relative state" formulation of quantum mechanics, Rev. Mod. Phys. **29**, 454 (1957).

[5] C. Rovelli, Relational quantum mechanics, Int. J. Theor. Phys. **35**, 1637 (1996).

[6] D. Bohm, A suggested interpretation of the quantum theory in terms of "hidden" variables I and II, Phys. Rev. **85**, 166, 180 (1952).

[7] E. Joos and H. D. Zeh, The emergence of classical properties through interaction with the environment, Z. Phys. B **59**, 223 (1985).

[8] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, Rev. Mod. Phys. **75**, 715 (2003).

[9] W. G. Unruh, Notes on black-hole evaporation, Phys. Rev. D **14**, 870 (1976).

[10] H. Umegaki, Conditional expectation in an operator algebra IV (entropy and information), Kōdai Math. Sem. Rep. **14**, 59 (1962).

[11] G. Lindblad, Completely positive maps and entropy inequalities, Commun. Math. Phys. **40**, 147 (1975).

[12] R. Haag, *Local Quantum Physics*, 2nd ed. (Springer, Berlin, 1996).

[13] E. H. Lieb and D. W. Robinson, The finite group velocity of quantum spin systems, Commun. Math. Phys. **28**, 251 (1972).

[14] M. Cheneau et al., Light-cone-like spreading of correlations in a quantum many-body system, Nature **481**, 484 (2012).

[15] X. Mi et al., Information scrambling in quantum circuits, Science **374**, 1479 (2021).

[16] Z. K. Minev et al., To catch and reverse a quantum jump mid-flight, Nature **570**, 200 (2019).

[17] R. Pazourek, S. Nagele, and J. Burgdörfer, Attosecond chronoscopy of photoemission, Rev. Mod. Phys. **87**, 765 (2015).

[18] M. O. Scully and K. Drühl, Quantum eraser: A proposed photon correlation experiment concerning observation and "delayed choice" in quantum mechanics, Phys. Rev. A **25**, 2208 (1982).

[19] J. F. Sandeman, Structural Zero Cosmological Constant and the Hubble Tension from Causal Actualization Density, in preparation (2026).

[20] J. F. Sandeman, Relational Actualism: Irreversible Events as the Primitive Basis of Physical Reality, in preparation (2026).